FREIGHT RATE PREDICTION

Model validation and December forecast

VALIDATION DESIGN

Chronological split: 43,147 rows before 2025-10 for fitting; 4,853 rows from 2025-10 held out. This simulates forecasting future loads and avoids random-split leakage across time.

MODEL AND DATA HANDLING

HistGradientBoostingRegressor predicts posted rate per mile; predictions are multiplied by distance. Features include endpoints, coordinates, equipment, weight, market/quote signals, and calendar features. Numeric missing values are handled natively; unseen cities map to an explicit unknown category. Load IDs are excluded from model features.

The fixed December chart inputs omit market/quote signals, and labeled data ends in October. The forecast is therefore stable across December dates; no unsupported daily or seasonal trend is inferred.

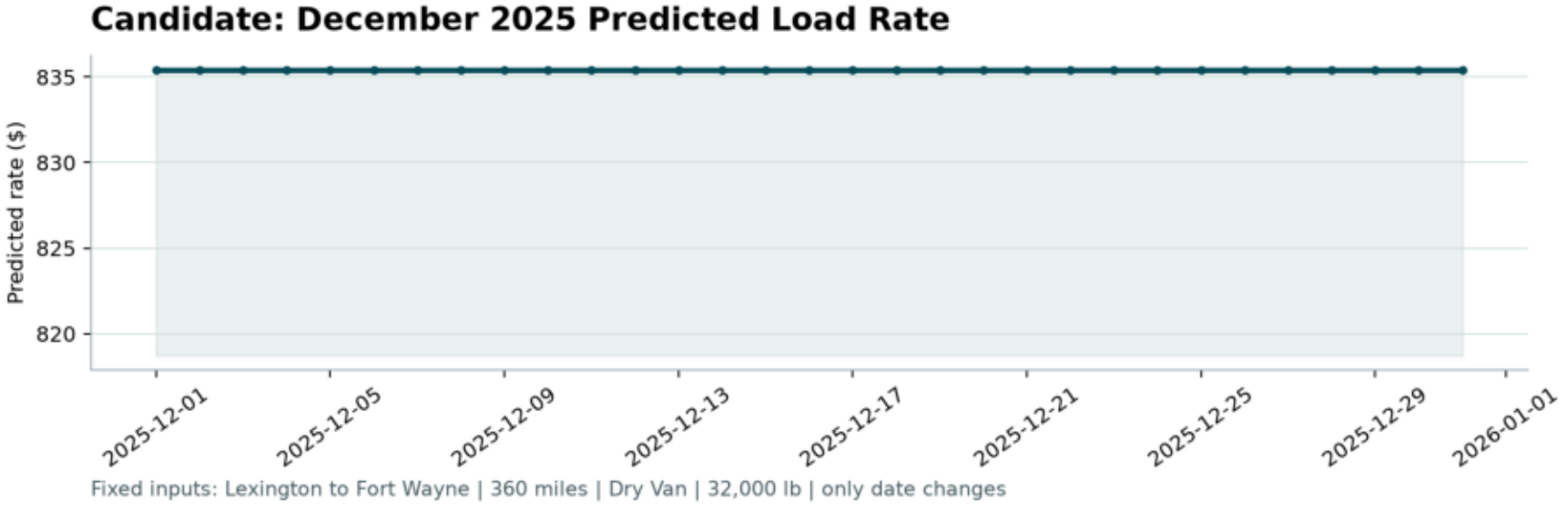
OCTOBER HOLDOUT RESULTS

MAE: $145.26

RMSE: $653.18

R^2: 0.8174

These metrics describe the October holdout only. Validation labels are not provided, so hidden-set performance is not measurable locally.


Candidate: December 2025 Predicted Load Rate
Predicted rate ($)
835
830
825
820
2025-12-01
2025-12-05
2025-12-09
2025-12-13
2025-12-17
2025-12-21
2025-12-25
2025-12-29
2026-01-01
Fixed inputs: Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lb | only date changes